NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

RACE Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2023
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED **MATHEMATICS** **M.M. 67**

REVISION ASSINGMENT # 14 (INDEFINITE AND DEFINITE INTEGRATION-02) TIME : 60 MIN.

SECTION–I

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. $\int \log(\sin^2 x.\cos x).\frac{(2-3\sin^2 x)}{\sin 2x}dx$ equals -

(A) $\frac{1}{4}\log^2\left(\frac{1}{2}\sin 2x.\sin x\right)+C$ (B) $\log\left(\frac{1}{2}\sin 2x.\sin x\right)+C$

(C) $\log^2\left(\sin 2x.\sin x\right)+C$ (D) $\log(\sin 2x.\sin x)+C$

(where C is the constant of integration)

2. If $\int \ell n\left(\sqrt{2+x}+\sqrt{2-x}\right)dx = x\ell n\left(\sqrt{2+x}+\sqrt{2-x}\right)+A\sin^{-1}\frac{x}{2}+Bx+C$, then $|B|$ is equal to

(A) $\frac{1}{2}$ (B) 1 (C) 2 (D) 3

(where C is constant of integration)

3. If $f(x)=\max\left\{x^2,(1-x)^2,\frac{3}{4}\right\}, x\in[0,1]$, then the value of $\int_0^1 f(x)dx$ is

Rankers Academy JEE

(A) $\frac{6\sqrt{3}-1}{12}$ (B) $\frac{3}{4}\left(\sqrt{3}-1\right)$ (C) $\frac{2-\sqrt{3}}{2}$ (D) $\frac{8-3\sqrt{3}}{24}$

4. $\int_{\pi/3}^{\pi/6} \ell n\sin x dx - \int_{-\ell n2}^{\frac{1}{2}\ell n\frac{3}{4}} \sin^{-1}e^x dx$ is equal to

(A) $\frac{\pi}{6}\ell n\frac{2}{3}$ (B) $\frac{\pi}{6}\ell n\frac{3}{2}$ (C) $\frac{\pi}{3}\ell n\frac{3}{4}$ (D) $\pi\ell n\frac{4}{3}$

5. Given $\int_0^y e^t dt+\int_0^x f(t)dt = g(x)$, then $\frac{dy}{dx}$ is equal to (where $f(x)$ is continuous and g(x) is a differentiable function)

(A) $\frac{g'(x)-f(x)}{1+g(x)-\int_0^x f(t)dt}$ (B) $\frac{g'(x)-f(x)}{g(x)-\int_0^x f(t)dt}$ (C) $e^y(g'(x)-f(x))$ (D) $\frac{(g'(x)-f(x)).e^y}{g(x)-\int_0^x f(t)dt}$

6. $\int e^x\left(\frac{1}{\ell nx}-\frac{1}{(\ell nx)(\ell nx^x)}\right)dx$ is equal to-

(A) $\frac{e^x}{\ell nx}+C$ (B) $e^x\ell nx+C$ (C) $2e^x.\log_x e+C$ (D) $\frac{e^x}{(\ell nx)^2}+C$

(where C is constant of integration)

For More Material Join: @JEEAdvanced_2024

7. If $f(x)$ is satisfies $\lim\limits_{x\to 0}\dfrac{x\ell n(1+x)}{\int_0^x f(t)\sin(t)\,dt}=5$ and continuous, then $f(0)$ is equal to-

(A) $\dfrac{2}{5}$ (B) 10 (C) 5 (D) $\dfrac{5}{2}$

8. If $I_1=\int_0^1\dfrac{e^x}{1+x}dx$, $I_2=\int_0^1\dfrac{x^2}{e^{x^3}(2-x^3)}dx$, then $\dfrac{I_1}{I_2}$ is equal to

(A) 3e (B) $\dfrac{3}{e}$ (C) $\dfrac{1}{3e}$ (D) $\dfrac{e}{3}$

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

9. If $\int\sqrt{\sec^2 x+3}\,dx=\ell n\left|f(x)+\sqrt{4+f^2(x)}\right|+\sqrt{3}\sin^{-1}\left(\dfrac{k.g(x)}{2}\right)+C$, where C is an integration constant, then

(A) value of k is equal to $\sqrt{3}$ (B) $g(x)=\dfrac{f(x)}{\sqrt{1+f^2(x)}}$

(C) $g^2(x)+\left(\sqrt{1+f^2(x)}\right)^{-2}=1$ (D) $g^2(x)f^2(x)=f^2(x)-g^2(x)$

Rankers Academy JEE

10. If $\int\left(2x^6+15x^4+2x^2+3\right)\cos(2x)dx=f(x)\sin(2x)+g(x)\cos(2x)+K$, where $f(x)$ and $g(x)$ are polynomial of x such that $f(0)=1$, $g(-1)=-4$ and K is constant of integration, then

(A) $4f(1)\le 3g(1)$ (B) $\int_{\frac{-\pi}{3}}^{\frac{\pi}{3}} f(x)dx\neq 0$ (C) $g''(0)<10$ (D) $\dfrac{g(2)}{f(2)}>1$

11. Let $f(x-3)=f(x+3)$ for all $x\in R$ and $f(x)=\begin{cases}x & ;\ 0\le x<3\\ 6-x & ;\ 3\le x<6\end{cases}$, then

(A) $\int_0^{36} f(x)dx=54$ (B) $\int_0^{48} f(x)dx=64$ (C) $\int_0^{54} f(x)dx=81$ (D) $\int_0^{18} f(x)dx=9$

12. Let $f(x)$ is differentiable bijective function such that $f'(x)-\int_0^x f(t)\,dt=\int_1^{f(x)} g(t)\,dt\ \forall x>0, f(0)=1$ and $g(x)=f^{-1}(x)$, then-

(A) $g(e)=\sqrt{2}$ (B) $g(1)=\sqrt{e}$ (C) $f(1)=\sqrt{e}$ (D) $f(e)=\sqrt{2}$

Space for Rough Work

For More Material Join: @JEEAdvanced_2024

Matching list type (4 × 4) (Single option correct) (3 Marks each, –1 for wrong answer)

13. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		**List-II**	
(P)	$\lim\limits_{n\to\infty}\left[\int\limits_0^2\left(1+\frac{t}{n+1}\right)^n dt\right]$ equals to	(1)	$e-\frac{1}{2}e^2-\frac{3}{2}$
(Q)	Let $f(x)$ be a function satisfying $f'(x) = f(x)$ with $f(0) = 1$ and g be the function satisfying $f(x) + g(x) = x^2$, then value of $\int\limits_0^1 f(x)g(x)dx$ is	(2)	e^2
(R)	$\int\limits_0^1 e^{(e^x)}(1+xe^x)dx$ is equals to	(3)	$e^2 - 1$
(S)	$\lim\limits_{k\to0}\frac{1}{k}\int\limits_0^k(1+\sin 2x)^{\frac{1}{x}}dx$, is equals to	(4)	e^e

Codes :

	P	Q	R	S
(A)	4	2	3	1
(B)	3	1	4	2
(C)	4	1	2	3
(D)	1	2	3	4

Rankers Academy JEE

Space for Rough Work

For More Material Join: @JEEAdvanced_2024

SECTION–III

Numerical Grid Type (Single digit Ranging from 0 to 9) (4 Marks each, –1 for wrong answer)

1. If $f(x)$ is an even function and $\alpha = \int_{2017}^{2019} (x-2017)^2 f(x-2017)\,dx$ and

$\beta = \int_{-2019}^{-2015} (x+2017)^2 f(x+2017)\,dx$, then $\frac{\beta}{\alpha}$ is $(\alpha \neq 0)$

2. The value of integral $\int_0^4 \min\{|x-1|,|x-2|,|x-3|\}\,dx = \frac{a}{b}$ where a and b are co-prime, then the value of (a + b) is

3. If $\int_0^{10} f(x)\,dx = 9$, then value of $\sum_{r=1}^{10} \int_0^1 f(x+r-1)\,dx$ is

4. Let $f : R \to R$ be non-constant differentiable function and satisfies, $f(x) = x^2 - \int_0^1 (x + f(t))^2\,dt$. Then $f(4)$ is equal to

5. $\frac{1}{n\pi} \int_{-n}^{n} \frac{dx}{\sqrt{\{x\}\{-x\}}}$, $n \in N$ is equal to (where $\{.\}$ fractional part function)

Rankers Academy JEE

6. Value of $\frac{29\int_0^1 (1-x^4)^7\,dx}{4\int_0^1 (1-x^4)^6\,dx}$ is equal to

Space for Rough Work

REVISION ASSINGMENT # 13 **II & DI-01** **MATHEMATICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	D	B	D	A	D	A,B,C,D	A,B,C	A,B,C
	Q.	11									
	A.	A,C,D									
SECTION-III	Q.	1	2	3	4	5					
	A.	2	4	4	7	3					
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		Q	S	P	P		P,T	T	P,T	R	

For More Material Join: @JEEAdvanced_2024